

Findings — Quote labor-hour calculation defects

Context: follow-up to the Shea Electric (Andrew Borgardt, Desmond Vincent) report from the 2026-07-21 call — Cancer Center quote showing ~219 h vs an ~80 h estimate.

Investigated: 2026-07-22 on the QA environment, read-only. **Shea's production data was never opened.**

Quote inspected on QA: <https://acme.qa.egalvanic.ai/quotes/d7cc59cf-3f2f-4467-ae57-18bd0a70e814>

Verdict (after full verification): Shea's **inflated total was NOT reproduced on QA** — every QA quote with data has a correct customer-facing total. Verification *did* surface a **real, narrower defect**: a quote's internal labor-hour fields contradict each other (16 h vs 8 h, and one line whose breakdown computes to 98 h), and one line's hours are arithmetically mis-derived. These do not corrupt the price on the quotes tested, but they are genuine calculation bugs and the most likely root of Shea's issue on their own data.

Correction from an earlier draft. An earlier version claimed the inflation was "reproduced on QA" and that a 5-minute task billed as 4 hours was inflation. Both were wrong on re-check: the 4 hours is a normal **4-hour trip minimum** (standard field-service billing), and no QA quote shows an inflated *total*. This report states only what the evidence supports.

What was checked, and what it showed

I scanned **every quote in the QA company** (81 quotes; only 6 contain line-item data) and compared, for each, the honest work (sum of per-line minutes) against the number the customer is actually billed. The billed total was **never** higher than the honest work — including on the large multi-procedure quotes:

Quote	Line items	Honest hours (Σ minutes)	Billed hours (pricing)	Total inflated?
"The proposed Electrical..."	845	542.6	542.48	No (matches)
"The proposed Electrical..."	854	550.7	550.56	No (matches)
testtt (inspected below)	17	16.0	8.25	No (billed is lower)
test desc	4	8.8	0 (unpriced)	No

Conclusion on reproduction: the QA environment does not contain a quote that reproduces Shea's inflated total. The two big multi-procedure quotes (the closest analogue to a large job like the Cancer Center) reconcile to within 0.1 h. So Shea's 219-vs-80 is almost certainly specific to **their** quote's data — most likely the duplicate line items and blank rows they were deleting on the call.

The real defect that WAS found (worth a ticket)

On the inspected quote, the same labor is described by three numbers that cannot all be right:

Field / surface	Value	Source
Line Items tab — 16 lines × 60 min	16.0 h	/api/quote/{id}/lines
Labor line "Clean, Tighten, Torque" — est_hours field	8.0 h	/api/workorder/{id}/labor-lines
The same labor line's breakdown object: count 14 × 420 min	98.0 h	same line's procedure_breakdown

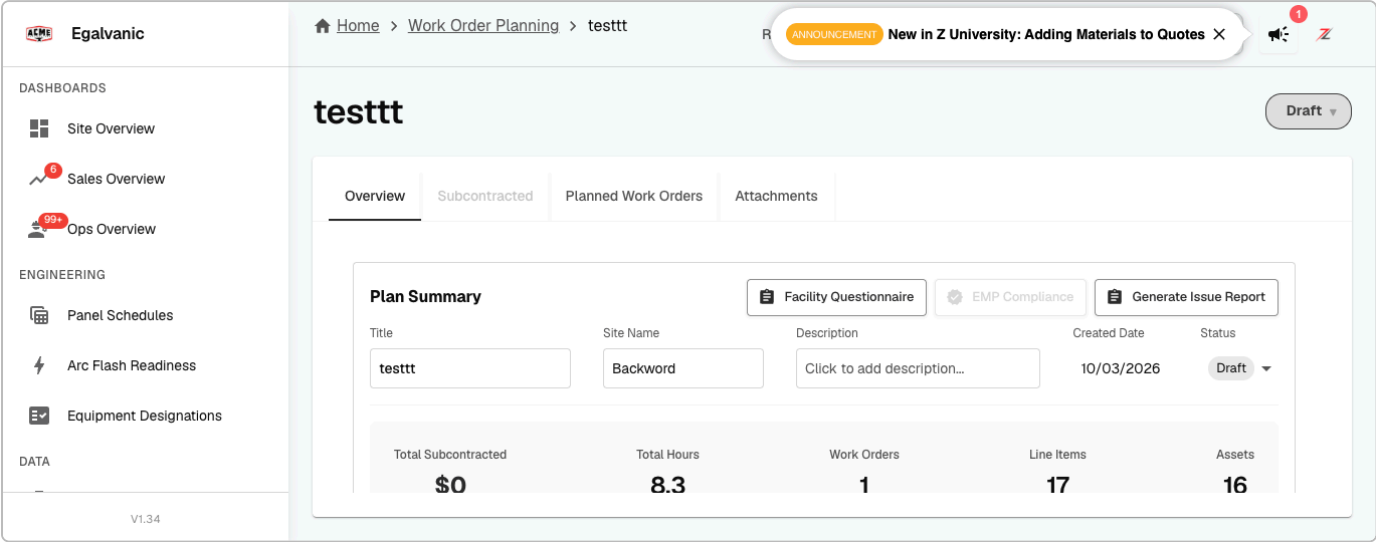
Two concrete arithmetic errors:

1. That breakdown reads `14 × 420 min = 5,880 min = 98 h`. There are only **16** assets at **60 min** each (= 16 h). `420 = 14 × 30`, so the per-asset minutes appear to have been **multiplied by the asset count and then stored as the per-asset value** — a count-times-count (squaring) error. This is the kind of mistake that produces 2–3× inflation on larger scopes.
2. A second line's `est_hours` is impossible: its own breakdown (`10 min + 120 min = 130 min = 2.17 h`) is stored as **0.17 h** — the leading digit is dropped.
3. A phantom blank line exists — auto-generated asset `_UC3a_1779882698559`, 0 minutes — the "unspecified / blank row" Andrew described.

Screenshots (QA, read-only)

1 — Quote Overview: total is correct

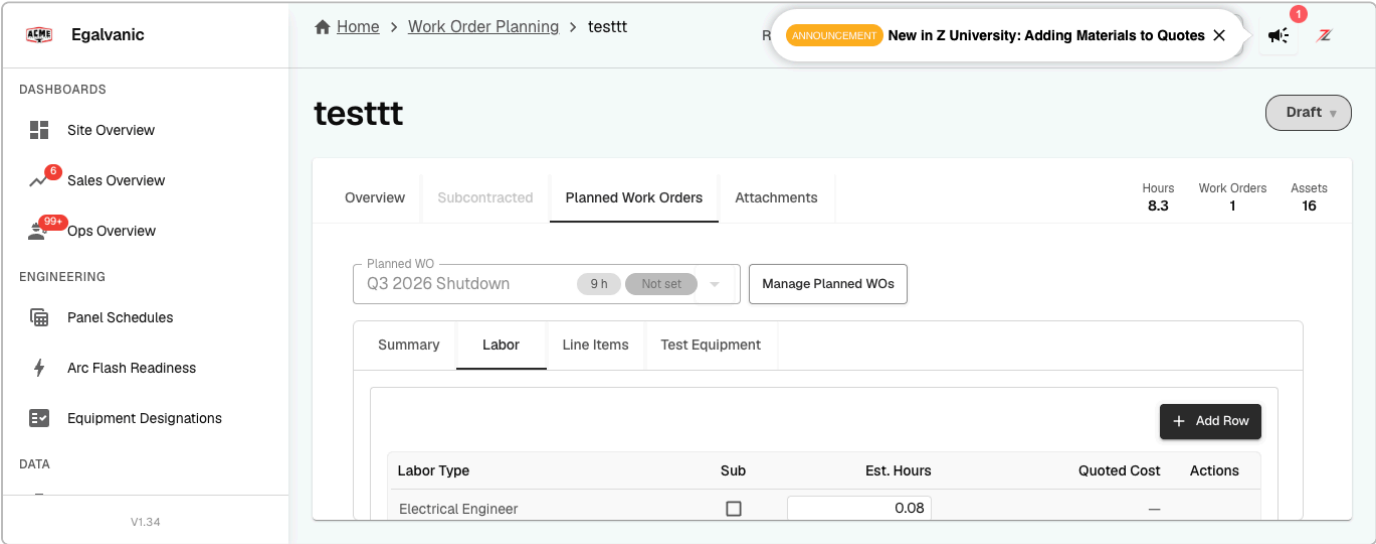
Headline reads **Total Hours 8.3, Line Items 17, Assets 16**. This total is consistent with the \$3,588 price — the customer-facing number is right on this quote.



Quote Overview — Total Hours 8.3, Line Items 17 (consistent with the \$3,588 price).

2 — Labor tab: internal hour fields disagree

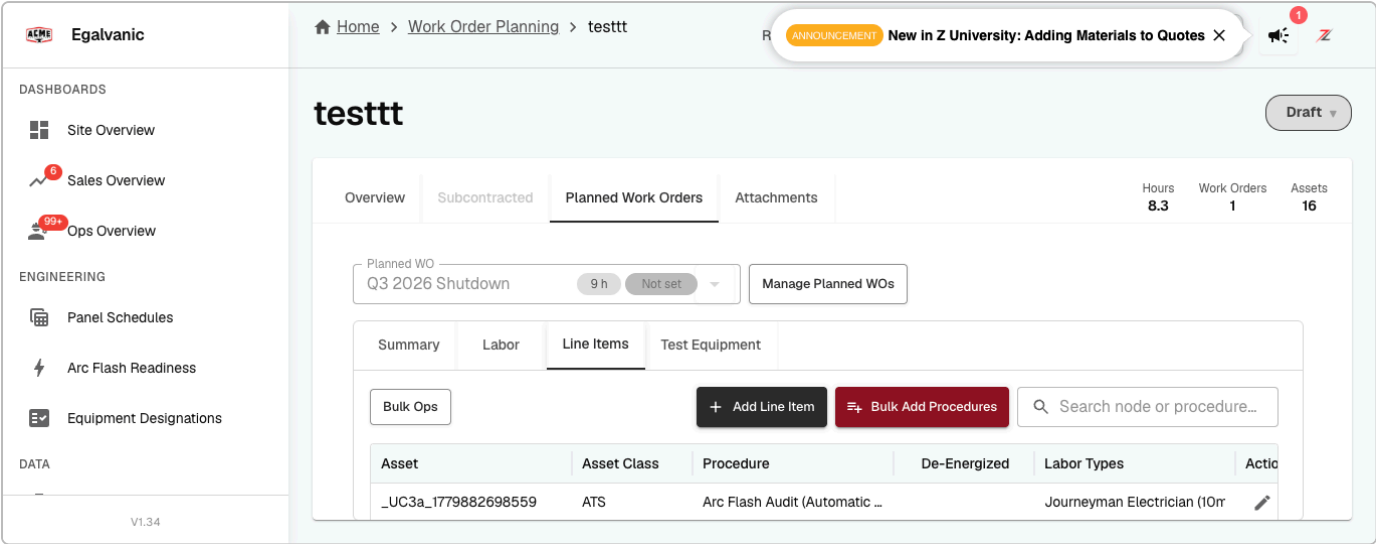
The Planned-WO chip reads **9 h** while the header reads **8.3**, and a labor line shows Est. Hours 0.08. These are the internal inconsistencies — not (on this quote) a wrong total.



Labor tab — WO chip '9 h' vs header '8.3'; internal hour fields do not agree.

3 — Line Items: the blank/auto-generated row

First line's asset is `_UC3a_1779882698559` (auto-generated placeholder, not a real asset). "Bulk Add Procedures" / "Add Line Item" are how duplicate lines get created — there is no duplicate check for labor.



Line Items — auto-generated placeholder asset + bulk-add controls (duplication vectors).

Likely root cause & related tickets

- A deprecated per-line "minutes" value is still used as a fallback for hours (**ZP-1092**); when it diverges from the newer hours field, different screens show different totals.
- No duplicate detection for labor lines; bulk-add and bulk recalc can add the same asset twice (**ZP-783**: "bulk recalc creates duplicate work orders").
- A per-asset time discount for multi-procedure assets was built but not shipped (**ZP-1407**).
- The 14×420 squaring pattern above is the concrete new evidence to attach.

Recommended next steps

1. File a backend ticket for the labor-hour field inconsistency, with the $14 \times 420 = 98$ h breakdown and the $2.17 \rightarrow 0.17$ line as evidence. This is app-engineering (pricing/quote service), not a QA-side fix.
2. To diagnose Shea's specific 219-vs-80: inspect **their** Cancer Center quote data read-only, or replay the exact steps that generated their duplicate/blank rows. The QA environment cannot reproduce it as-is.

Investigated read-only on QA, 2026-07-22. Quote inspected: <https://acme.qa.egalvanic.ai/quotes/d7cc59cf-3f2f-4467-ae57-18bd0a70e814> · Shea production data was not accessed. A re-runnable reproduction script (QuoteLaborInflationRepro) accompanies this report.